

MARCH, 2024

Mixed Reality Multi-user Asymmetric Telecollaboration

PHD ORAL DEFENCE



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Supervisors



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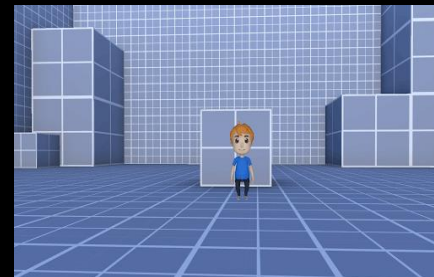
CMIC | Computational
Media
Innovation
Centre



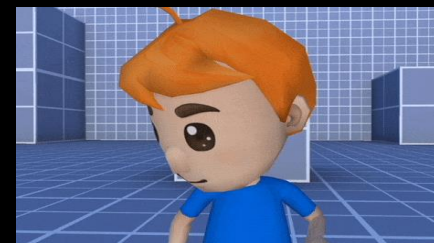
VICTORIA UNIVERSITY OF
WELLINGTON
TE HERENGA WAKA



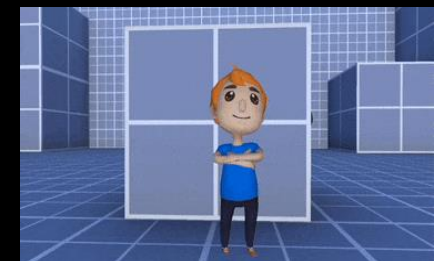
Collaborative Tools



Limited Field of view

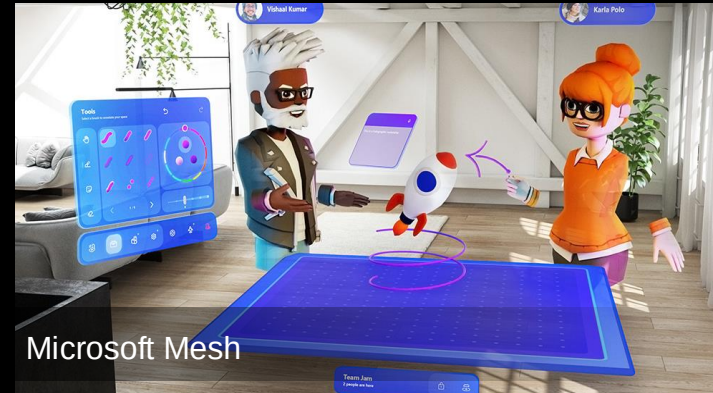


Barely see others' body language & gaze



Fixed viewpoint

Collaborative XR Tools



Research Problem



totally disconnected from the real world's physical space.



face tracking only work for one person at a time and setup is quite expensive.



remote users have limited physical presence and interaction.



users collaborate in a shared virtual space not in physical space.



shared physical space with sense of presence and tools to assist their task

Collaborate in Asymmetric Physical Space with VR and AR Users



Remote Users

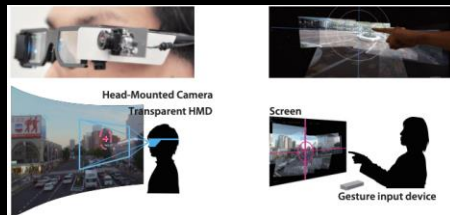
- VR systems are ideal for Strong storytelling
- Provide a fully immersive/3D experience
- Easily change the scale and representation of the virtual body



Local Users

- AR systems are ideal for live sharing
- Allow real world overlay/can share viewpoints
- Remote annotation/link support
- Improve the real-world task

Related Work



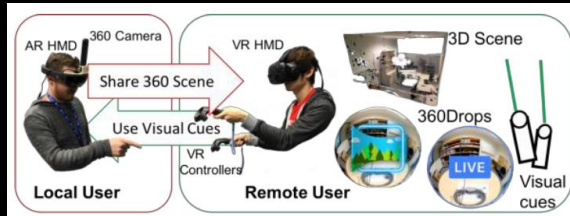
[S. Kasahara and J. Re kim oto 2014]



[S. Nagai et al. 2015]



[Lee et al. 2017]



[Teo et al. 2017]



[Lee et al. 2020]



[Rhee et al. 2020]

Only One-One & One-Many

No Audio-Visual Sync

No Multi-Viewpoints Management

Research Questions

How can multiple remote (VR) users effectively connect and collaborate with local (AR) users in the Mixed Reality Collaboration space?

teleportation

RQ1

How can **multiple remote users** have a **high sense of presence** while maintaining spatial awareness and understanding in the Mixed Reality Collaboration space?

representation

RQ2

How can **multiple remote users** be **effectively represented** in the Mixed Reality Collaboration space to achieve **co-presence**?

interaction

RQ3

How can **multiple users** **interact effectively** in the Mixed Reality Collaboration space?

view point perspective

RQ4

How can users' **viewpoints** be **managed effectively** to support various types of collaborative tasks in the Mixed Reality Collaboration space?

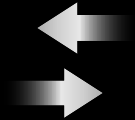
multi-perspective collaboration

RQ5

How can **multi-perspective** collaboration be **effectively** achieved for supporting collaborative tasks?

Contributions

1 Mixed Reality Multi-user Asymmetric Collaboration (**MRMAC**) system



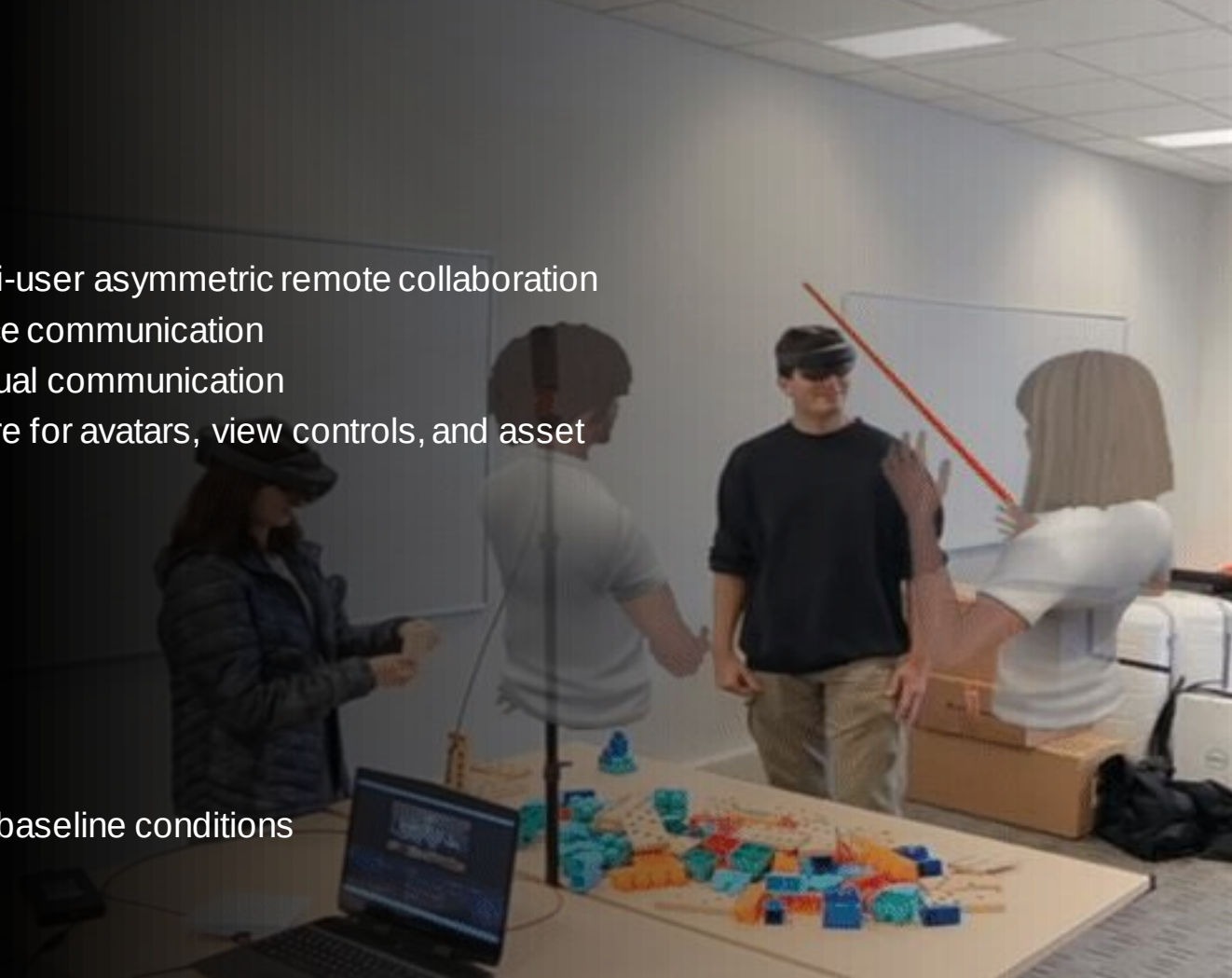
MRMAC

Design & Implementation

- Design concept for multi-user asymmetric remote collaboration
- Bidirectional face-to-face communication
- Synchronised audio-visual communication
- Client-server architecture for avatars, view controls, and asset streaming

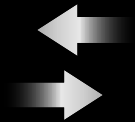
Evaluation

- System evaluation
 - scalability and latency
- User evaluation
 - impact of user roles
 - compared against two baseline conditions



Contributions

1 Mixed Reality Multi-user Asymmetric Collaboration (**MRMAC**) system



2 Vicarious, a context-aware viewpoint selection method



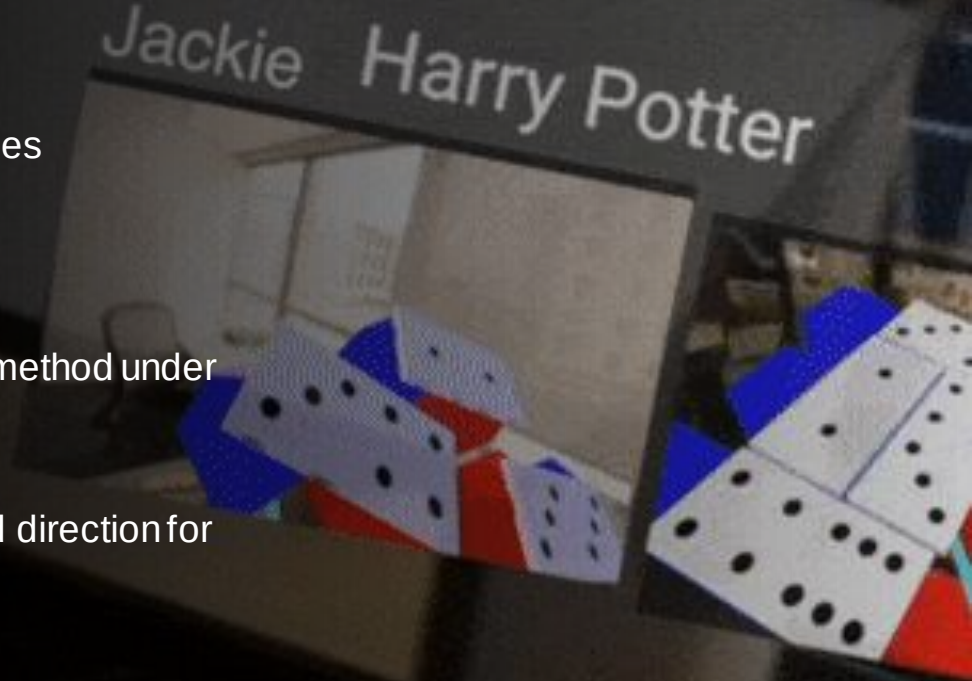
Vicarious

Design & Implementation

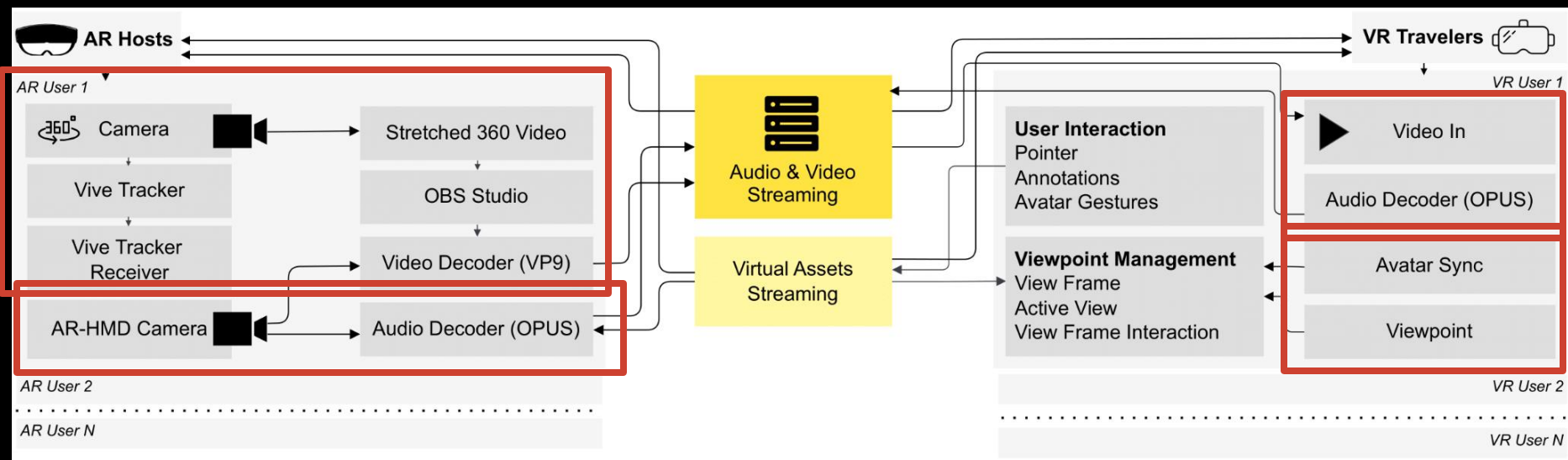
- Context-aware viewpoint selection method based on visual saliency, user actions, and speech patterns.
- Various visualization and interaction techniques

User Study

- Evaluated the impact of viewpoint selections method under four distinct conditions
- Recommendations for design implications and direction for future research



MRMAC Architecture



Communication Cues

Viewpoint Sharing



Multiple Avatar Control



Personalised Avatar

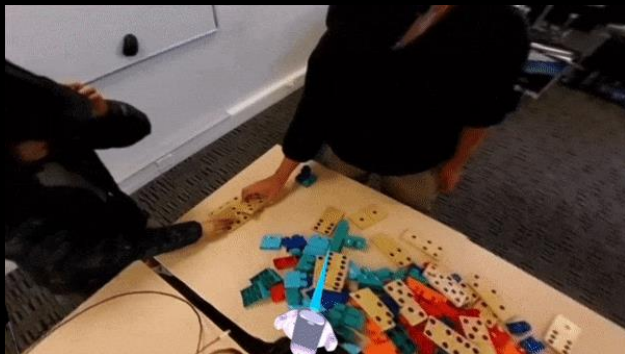


Spatial Audio

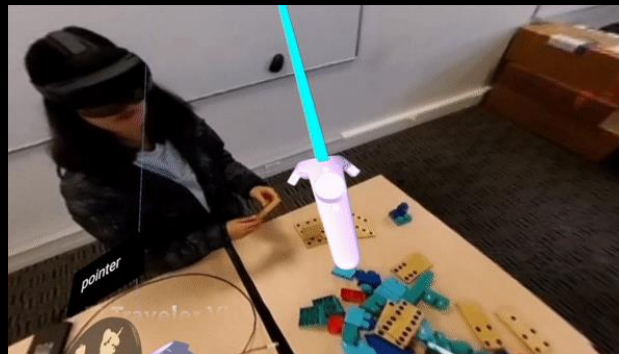


Interaction Cues

3D Pointer



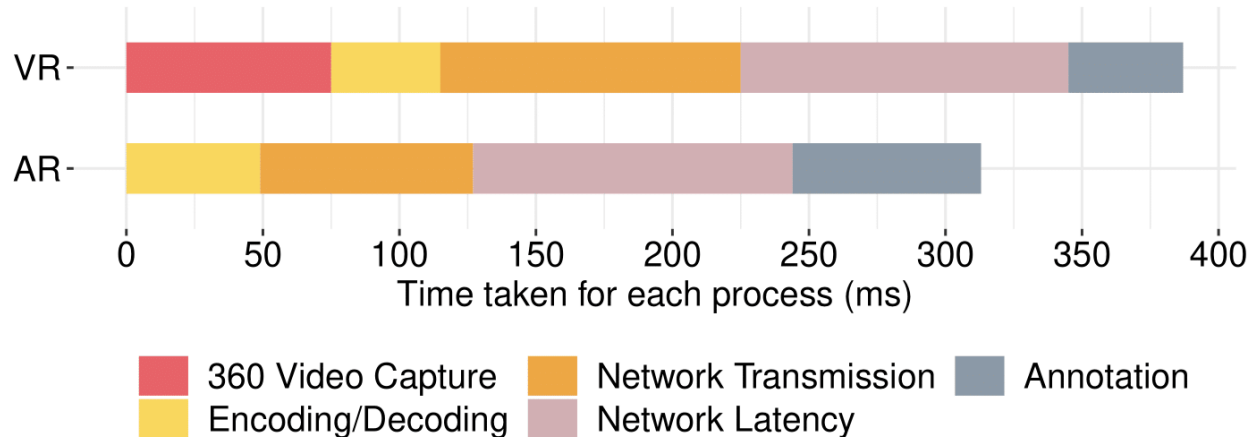
Annotation



System Evaluation: Latency

Average end-to-end latency

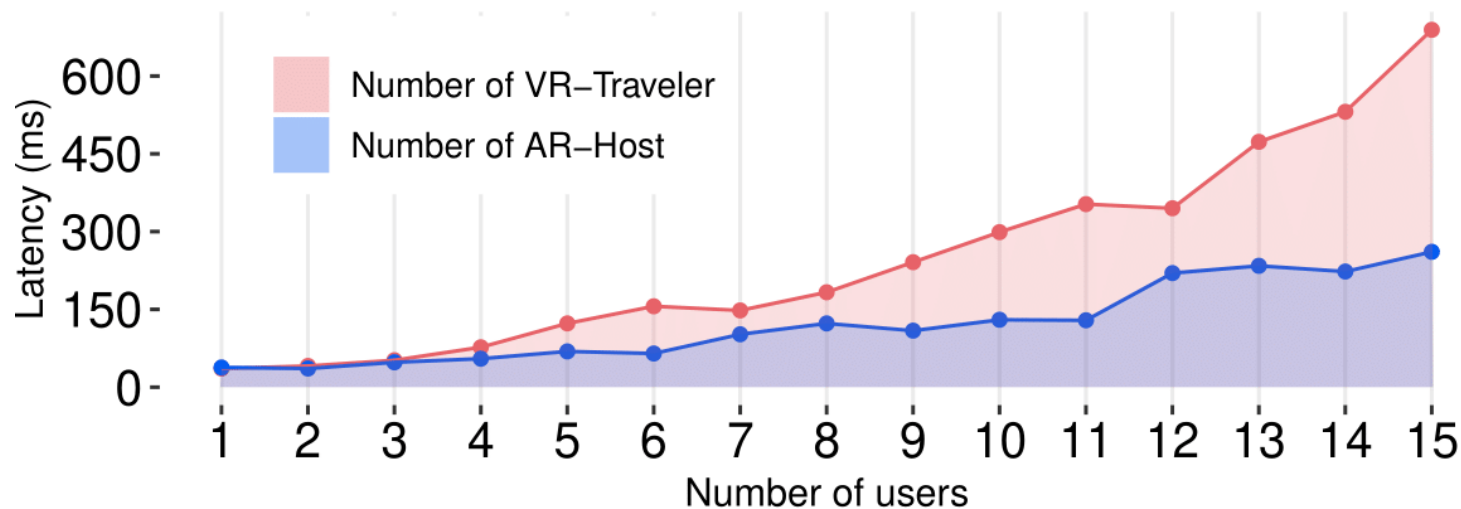
- Changes in the 360-degree video to show up on the local computer and in the VR/AR window.



System Evaluation: Scalability

Network latency as the number of users increases

- average frame rate = video streaming 30FPS
- audio streaming 44.1kHz
- rendering time of 60 ± 10 FPS.
- wireless bandwidth capacity (AX3000 Wi-Fi 6 router) 3 Gbps



User Evaluation

36 participants

18—81 years

$\mu = 30.83, \sigma = 14.10$

3×2 mixed factorial design

between-subjects factor:
local vs. remote

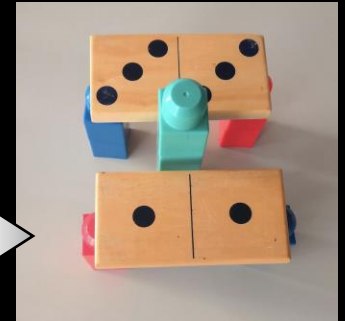
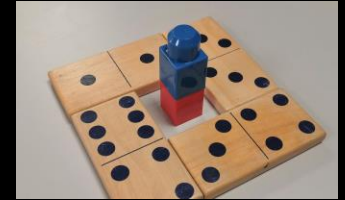
within-subjects factor:
C1 vs. C2 vs. C3

VR Users



Virtual replica

AR Users



Physical model

User Evaluation

3×2 mixed factorial design

between-subjects factor:
local vs. remote

within-subjects factor:
C1 vs. C2 vs. C3

1

Condition 1
Conventional video
with 2D annotation

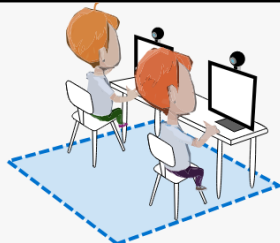
2

Condition 2
360 Video without
augmented visual cues

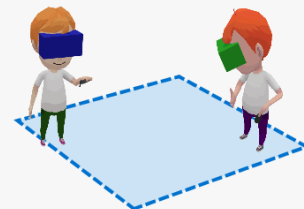
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Condition 3
360 video with
augmented visual cues

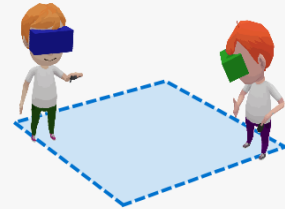
VR-Travelers



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
Mouse for 2D Annotation
Representation:
Appears in 2D video

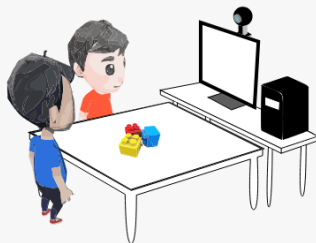


VR HMD (Viewing 360 Video)
Camera Control by VR HMD
N/A
No Visual, Voice Only

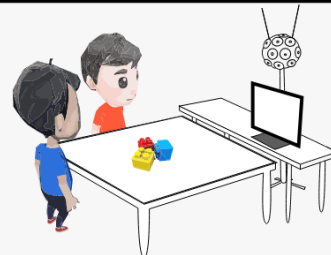


VR HMD (Viewing 360 Video)
Camera Control by VR HMD
VR Controller
3D Avatar

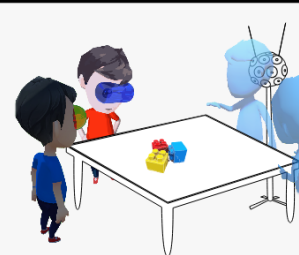
AR-Hosts



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
N/A
Representation:
Appears in 2D video



Desktop Monitor
N/A
N/A
Appears in 360-degree Video



AR HMD
N/A
N/A
Appears in 360-degree Video

User Evaluation

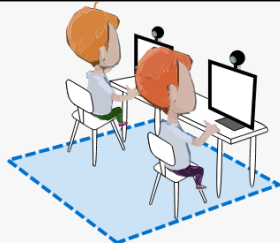
Condition 1



1

Condition 1
Conventional video
with 2D annotation

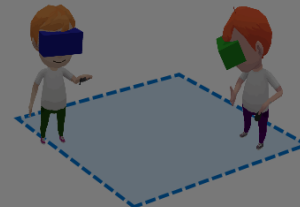
VR-Travelers



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
Mouse for 2D Annotation
Representation:
Appears in 2D video

2

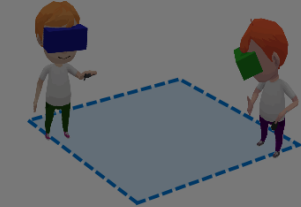
Condition 2
360 Video without
augmented visual cues



VR HMD (Viewing 360 Video)
Camera Control by VR HMD
N/A
No Visual, Voice Only

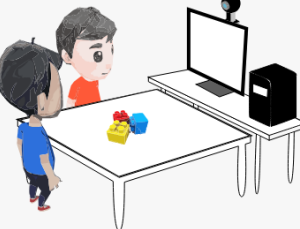
3

Condition 3
360 video with
augmented visual cues

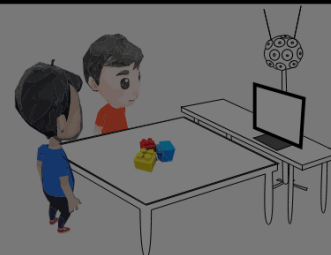


VR HMD (Viewing 360 Video)
Camera Control by VR HMD
VR Controller
3D Avatar

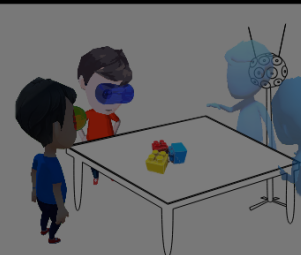
AR-Hosts



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
N/A
Representation:
Appears in 2D video



Desktop Monitor
N/A
N/A
Appears in 360-degree Video



AR HMD
N/A
N/A
Appears in 360-degree Video

User Evaluation

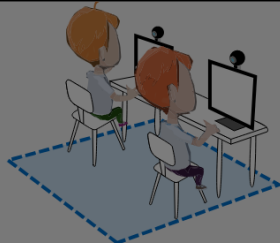
Condition 2



1

Condition 1
Conventional video
with 2D annotation

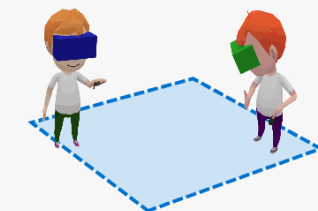
VR-Travelers



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
Mouse for 2D Annotation
Representation:
Appears in 2D video

2

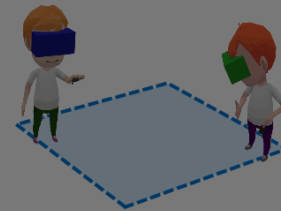
Condition 2
360 Video without
augmented visual cues



VR HMD (Viewing 360 Video)
Camera Control by VR HMD
N/A
No Visual, Voice Only

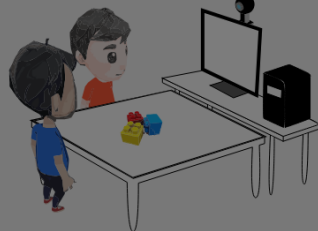
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Condition 3
360 video with
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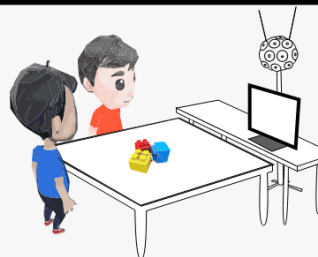


VR HMD (Viewing 360 Video)
Camera Control by VR HMD
VR Controller
3D Avatar

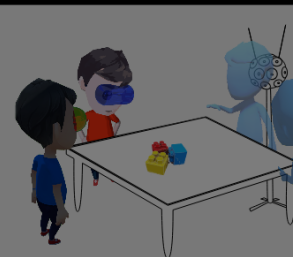
AR-Hosts



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
N/A
Representation:
Appears in 2D video



Desktop Monitor
N/A
N/A
Appears in 360-degree Video



AR HMD
N/A
N/A
Appears in 360-degree Video

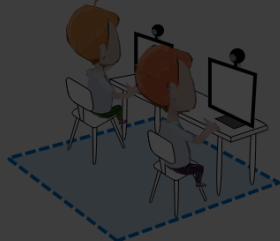
User Evaluation

Condition 3



1

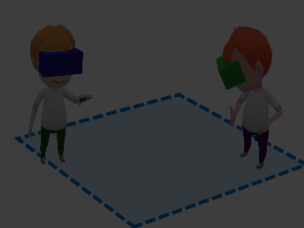
Condition 1
Conventional video
with 2D annotation



Viewing Device:
Desktop Monitor
Camera control:
N/A
Interaction:
Mouse for 2D Annotation
Representation:
Appears in 2D video

2

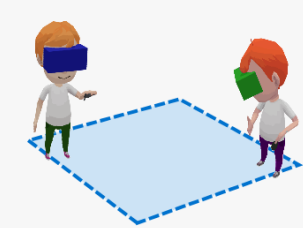
Condition 2
360 Video without
augmented visual cues



VR HMD (Viewing 360 Video)
Camera Control by VR HMD
N/A
No Visual, Voice Only

3

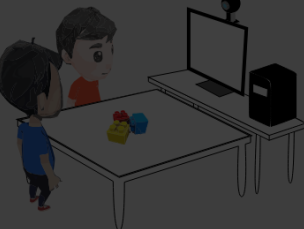
Condition 3
360 video with
augmented visual cues



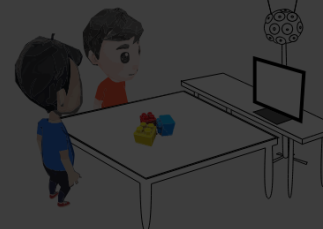
VR HMD (Viewing 360 Video)
Camera Control by VR HMD
VR Controller
3D Avatar

VR-Travelers

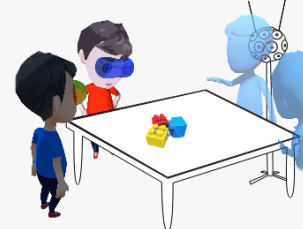
AR-Hosts



Viewing Device:
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Camera control:
N/A
Interaction:
N/A
Representation:
Appears in 2D video



Desktop Monitor
N/A
N/A
Appears in 360-degree Video



AR HMD
N/A
N/A
Appears in 360-degree Video

Results

Statistically Significant

- Spatial Presence
- Social Presence
- Task Workload
- System Usability
- Preference

Not Statistically Significant

- Task Completion Time
- Simulator Sickness

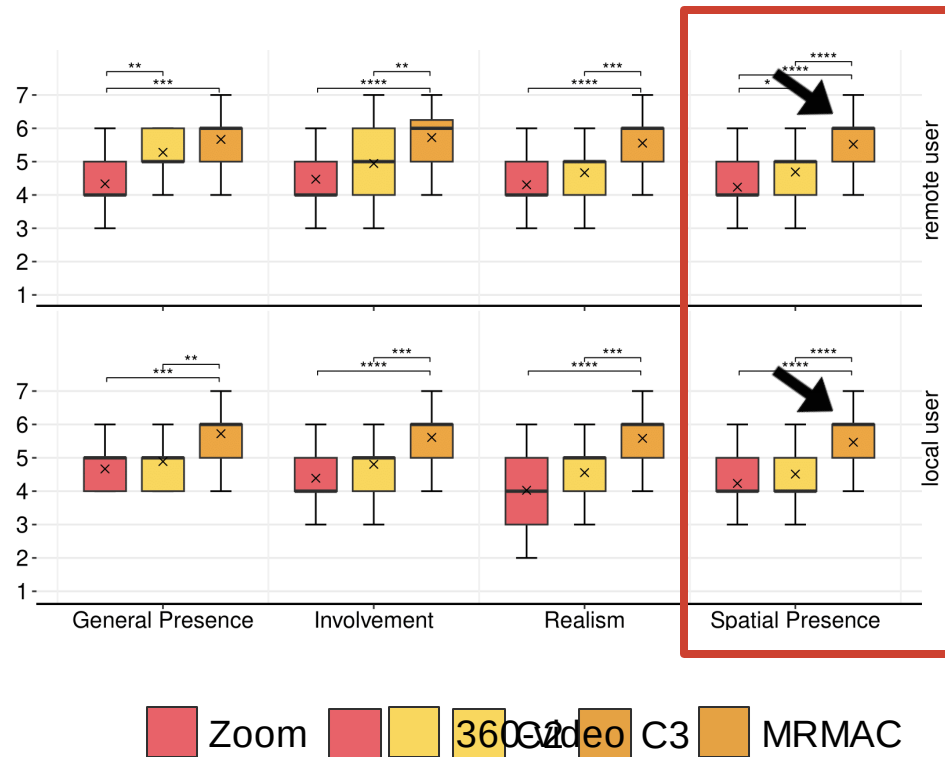
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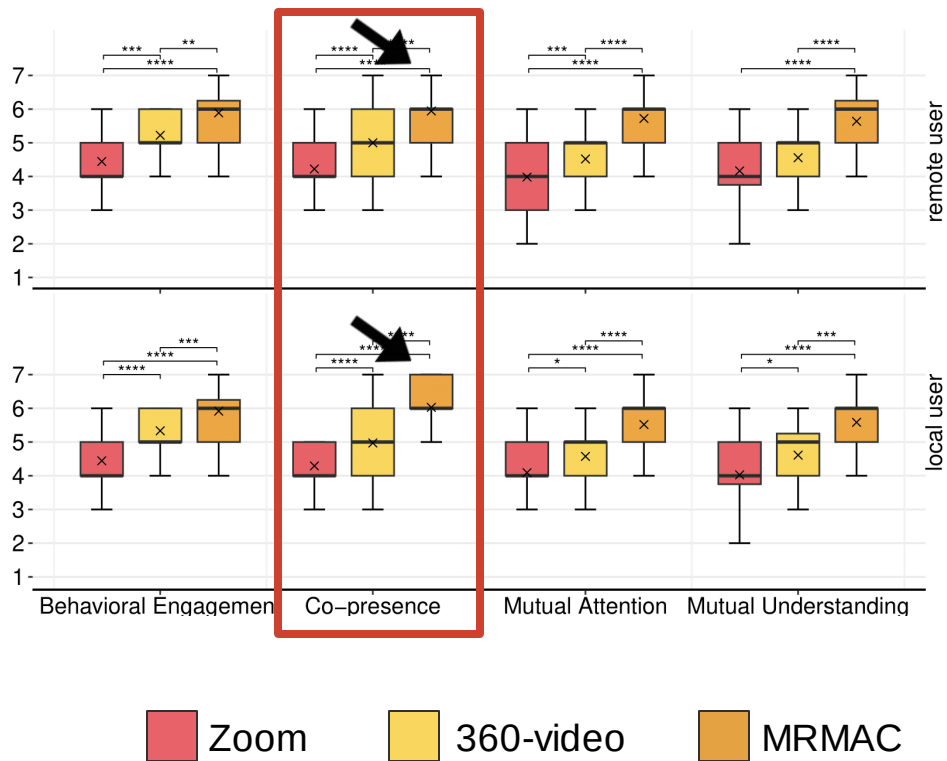
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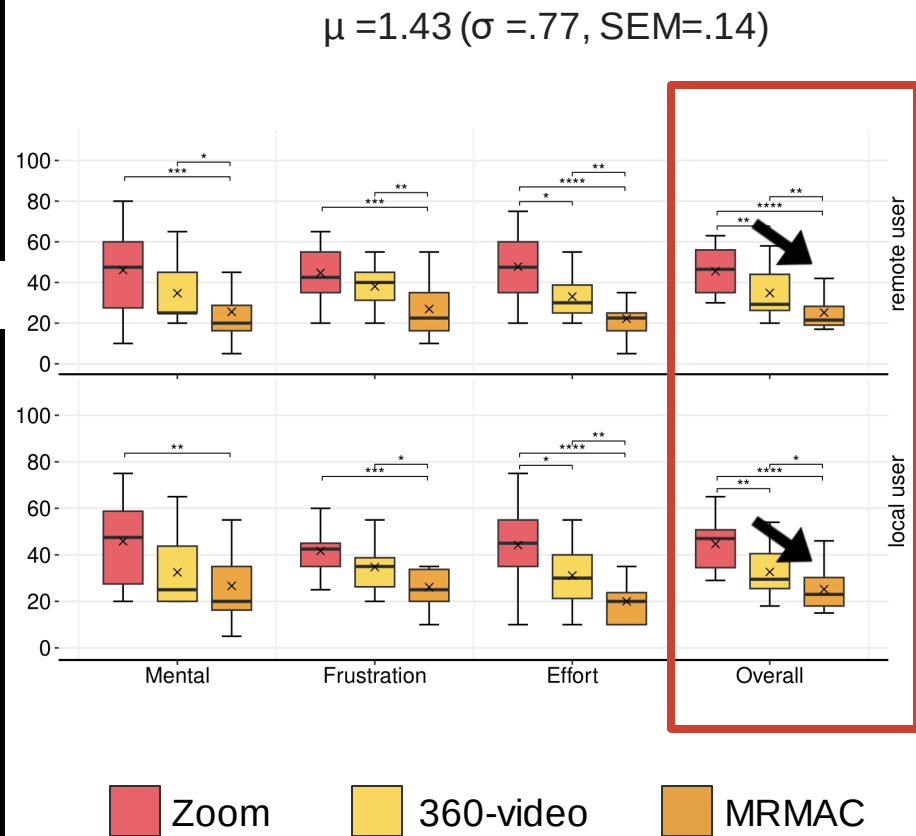
Results

Statistically Significant

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Not Statistically Significant

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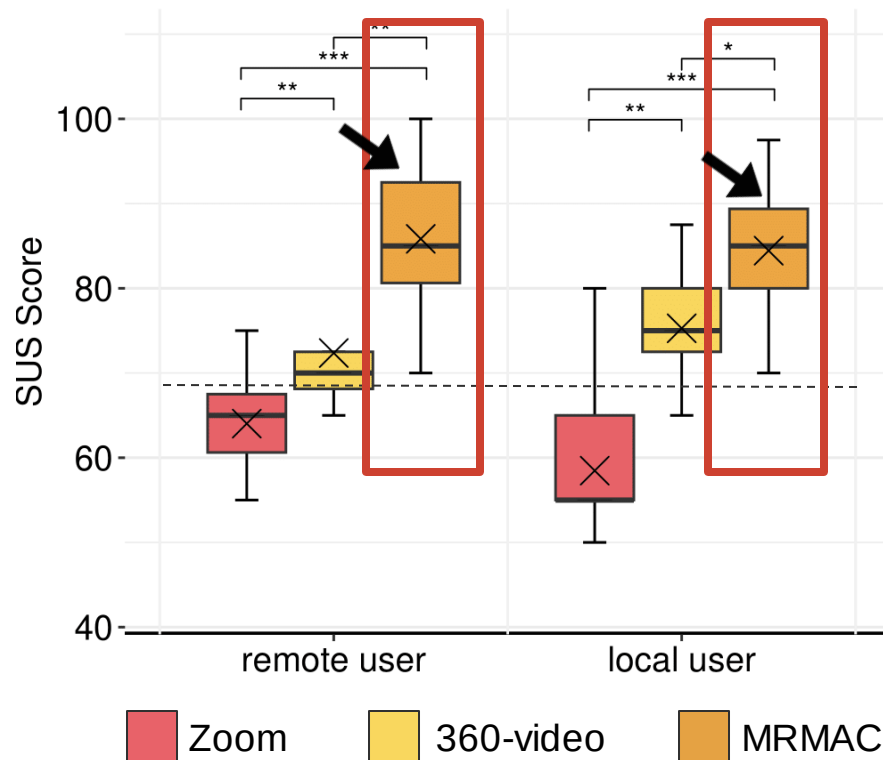
Results

Statistically Significant

- Spatial Presence
- Social Presence
- Task Workload
- **System Usability**
- Preference

Not Statistically Significant

- Task Completion Time
- Simulator Sickness



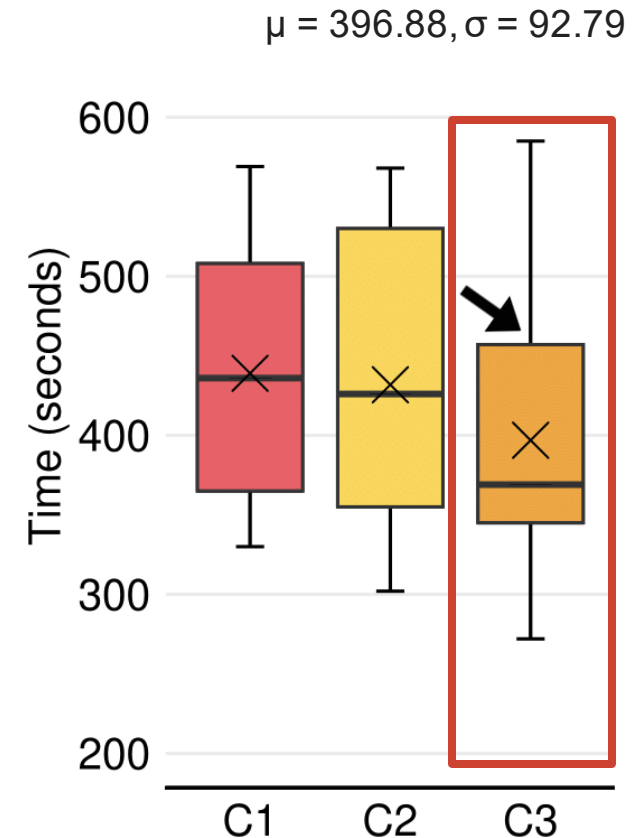
Results

Statistically Significant

- Spatial Presence
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- Preference

Not Statistically Significant

- Task Completion Time
- Simulator Sickness



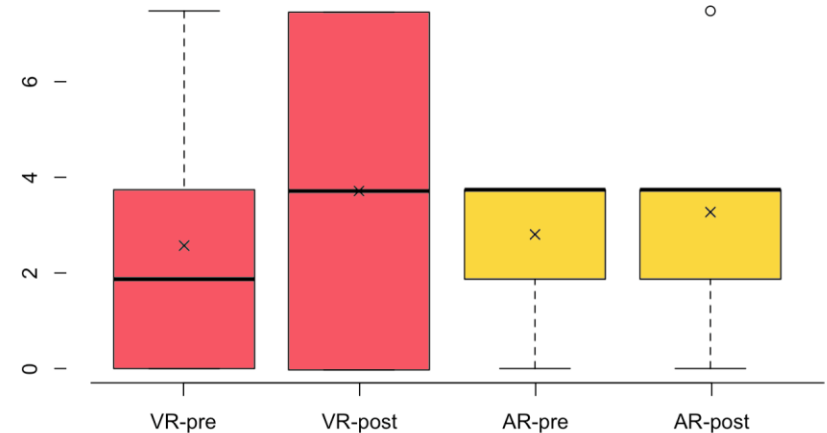
Results

Statistically Significant

- Spatial Presence
- Social Presence
- Task Workload
- System Usability
- Preference

Not Statistically Significant

- Task Completion Time
- Simulator Sickness



Summary of MRMAC

RQ1: teleportation

RQ2: representation

RQ3: interaction

Fully bidirectional asymmetric collaboration system

- Real-time collaboration
- Audio-visual synchronisation
- Virtual assets blending

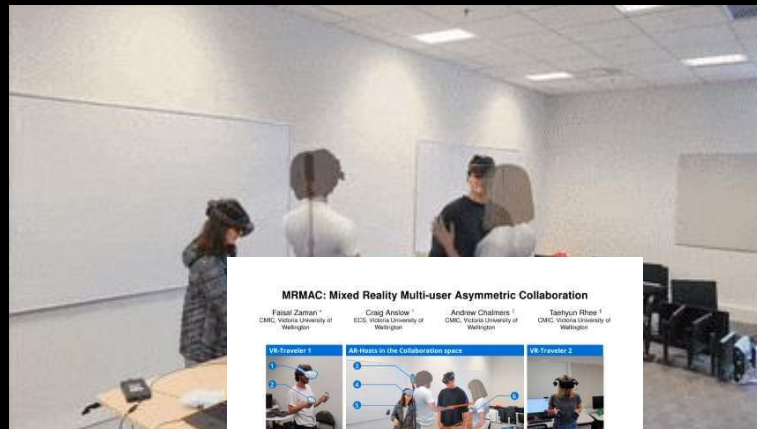
System evaluation

- Reasonably Scalable
- Low-latency

User evaluation

- MRMAC *performed* collaborative tasks faster
- MRMAC *preferred* over conventional 2D and traditional 360

Why travel with hefty backpack
when you can teleport with **MRMAC**



F. Zaman, C. Anslow, A. Chalmers, T. Rhee. **MRMAC: Mixed Reality Multiuser Asymmetric Collaboration**. IEEE ISMAR, 2023.

Multiple Viewpoints

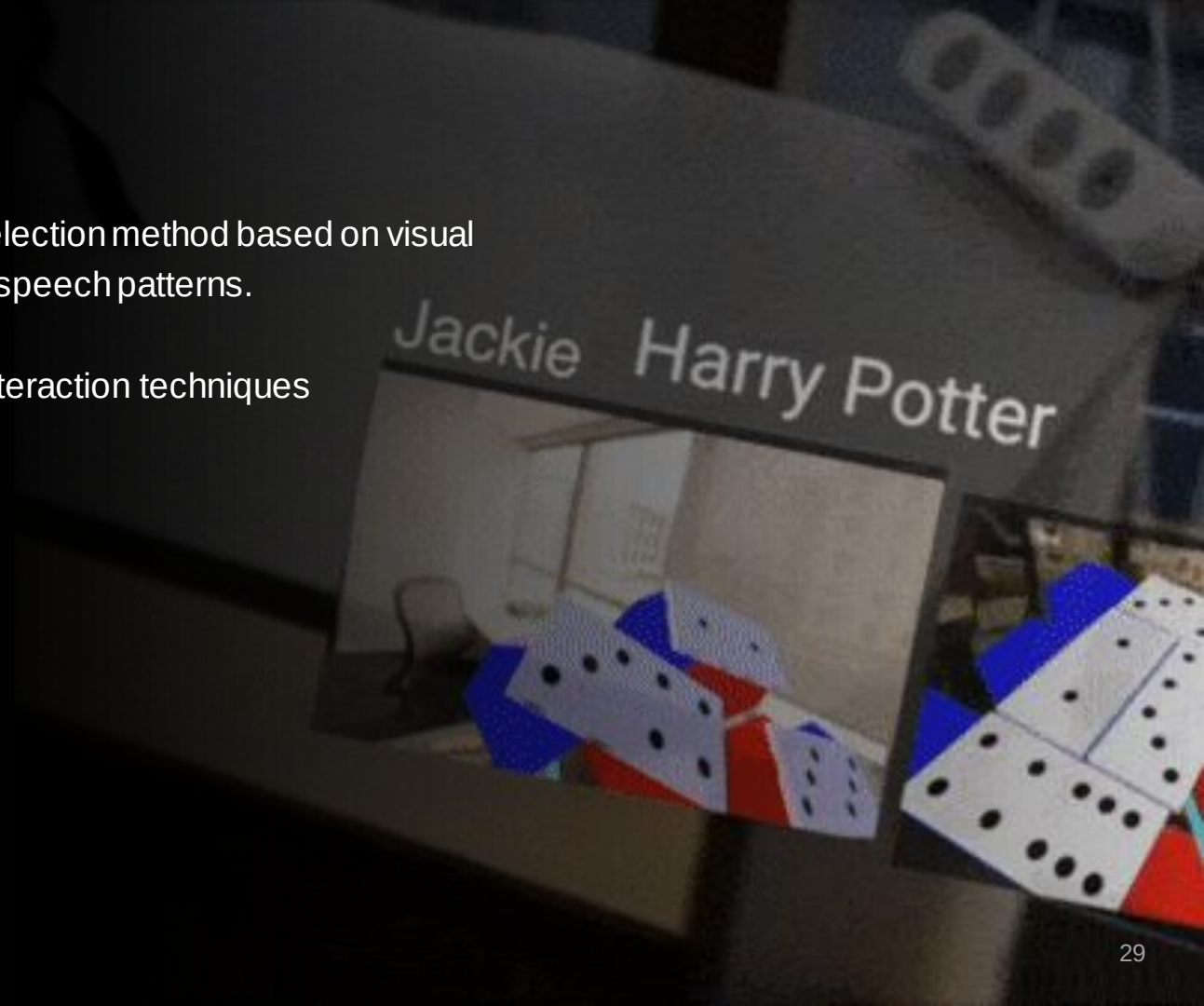
Mixed-perspective shown to improve the collaborative experience in remote settings

managing multiple viewpoints challenging for users to decide which views to focus on



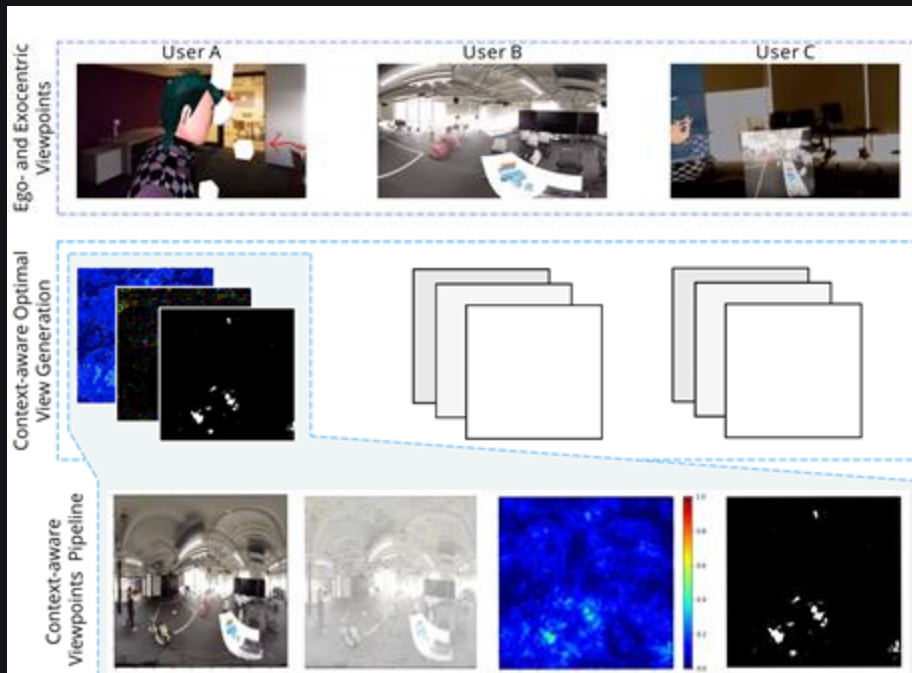
Vicarious

- Context-aware viewpoint selection method based on visual saliency, user actions, and speech patterns.
- Various visualization and interaction techniques



Viewpoint Selection

- Visual saliency



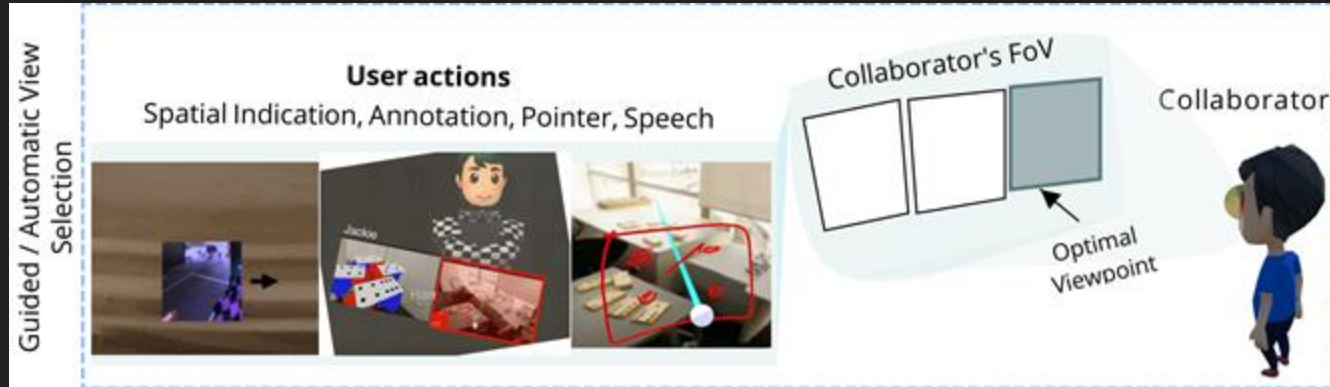
$$S(i) = w1 * K(i) + w2 * O(i) + w3 * G_{\sigma}(i)$$

Viewpoint Selection

Predefined actions

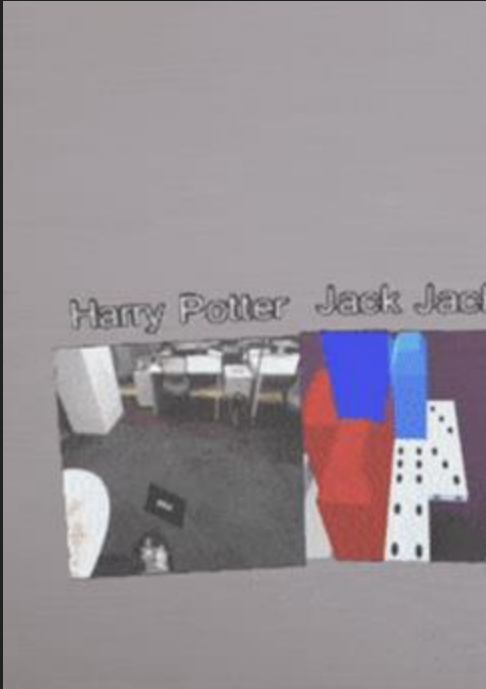
Verbal cues (*Voice activity + Intensity*)

Visual cues (*Point, Annotate, Gestures*)

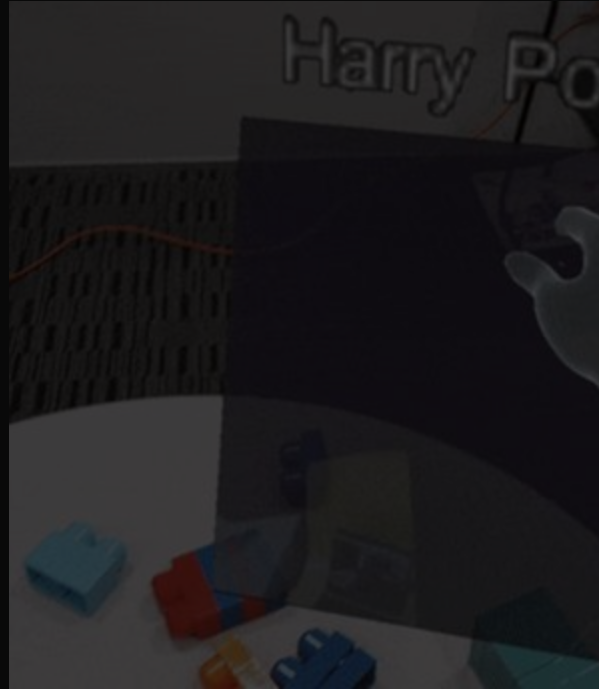


Visualization and Interaction

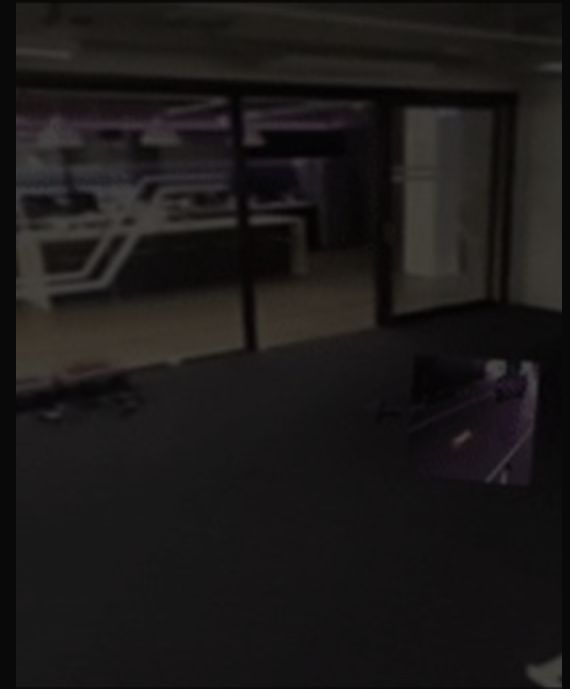
1. Smooth Follow



2. Pinned to Fixed Location

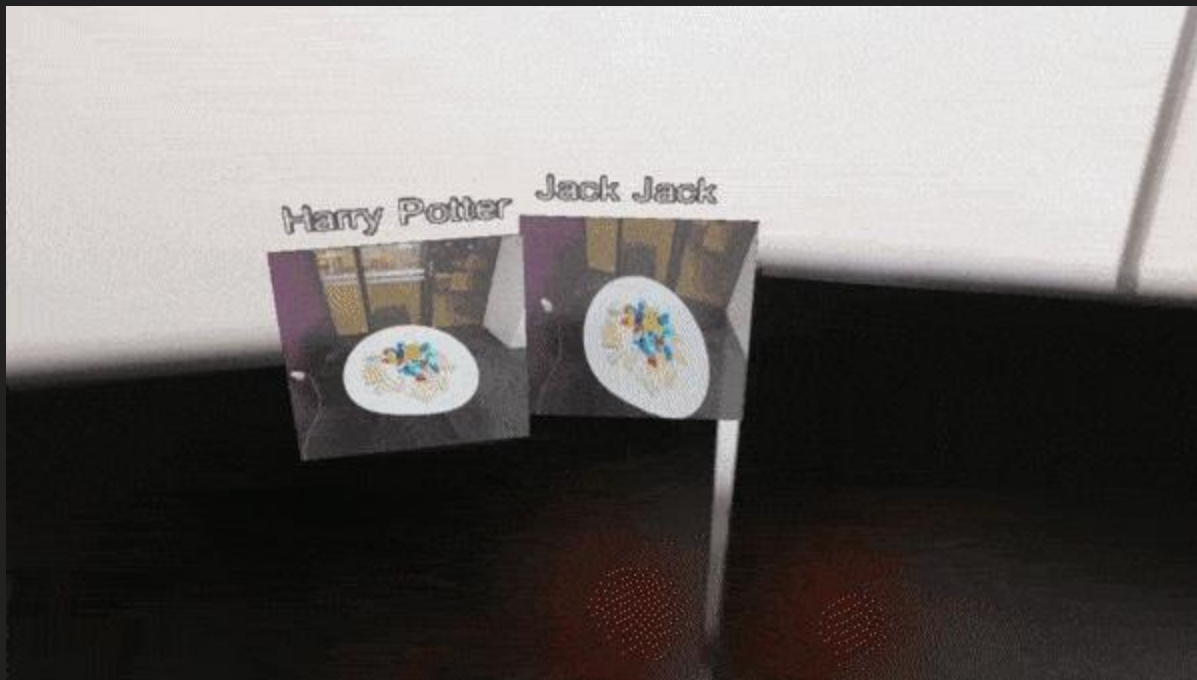


3. Spatial Indication



Visualization and Interaction

1. Smooth Follow



Visualization and Interaction

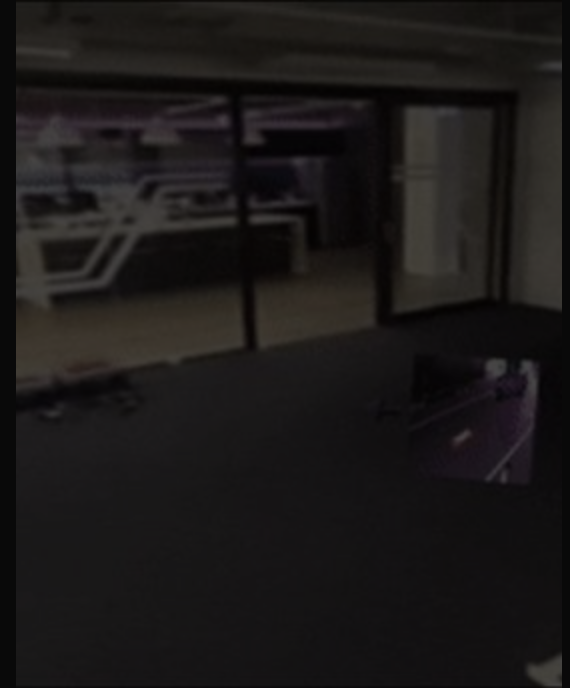
1. Smooth Follow



2. Pinned to Fixed Location



3. Spatial Indication



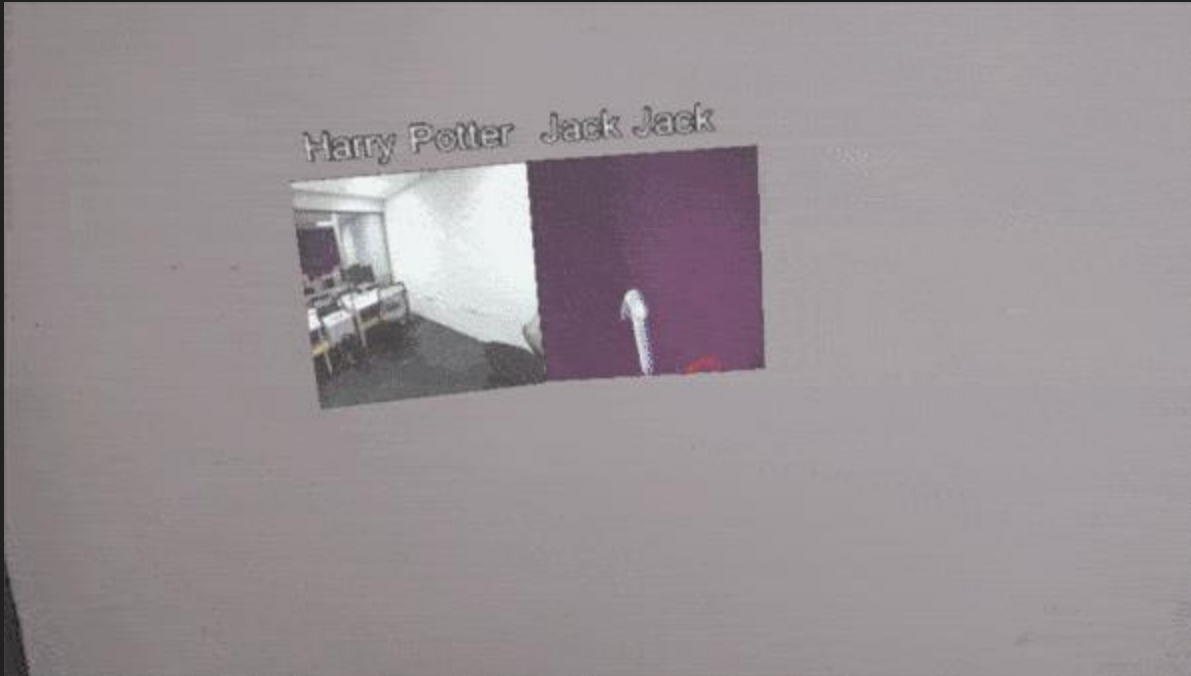
Visualization and Interaction

2. Pinned to Fixed Location



Visualization and Interaction

2. Pinned to Fixed Location (Zoom In/Out)

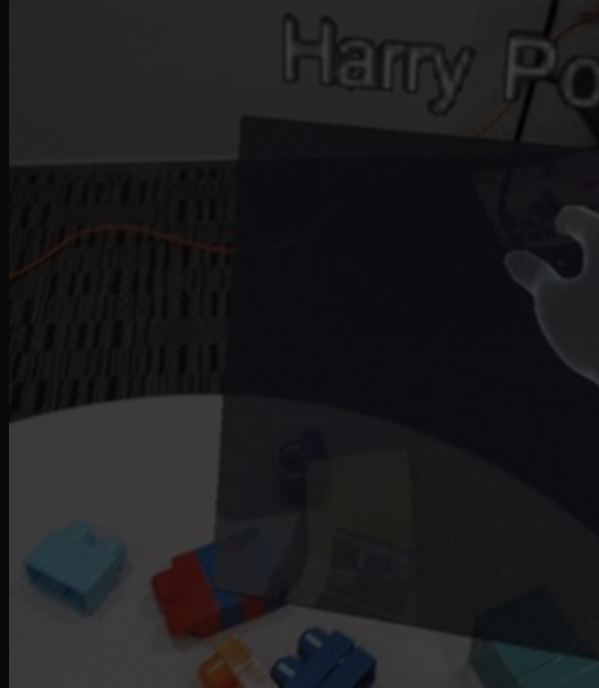


Visualization and Interaction

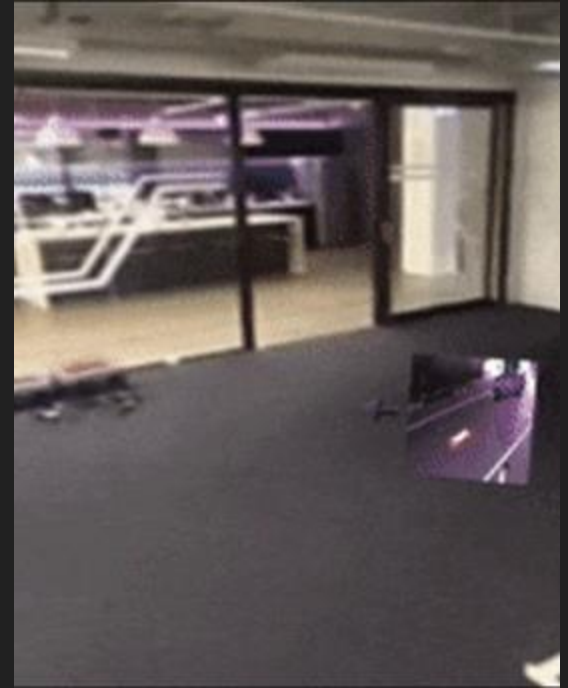
1. Smooth Follow



2. Pinned to Fixed Location

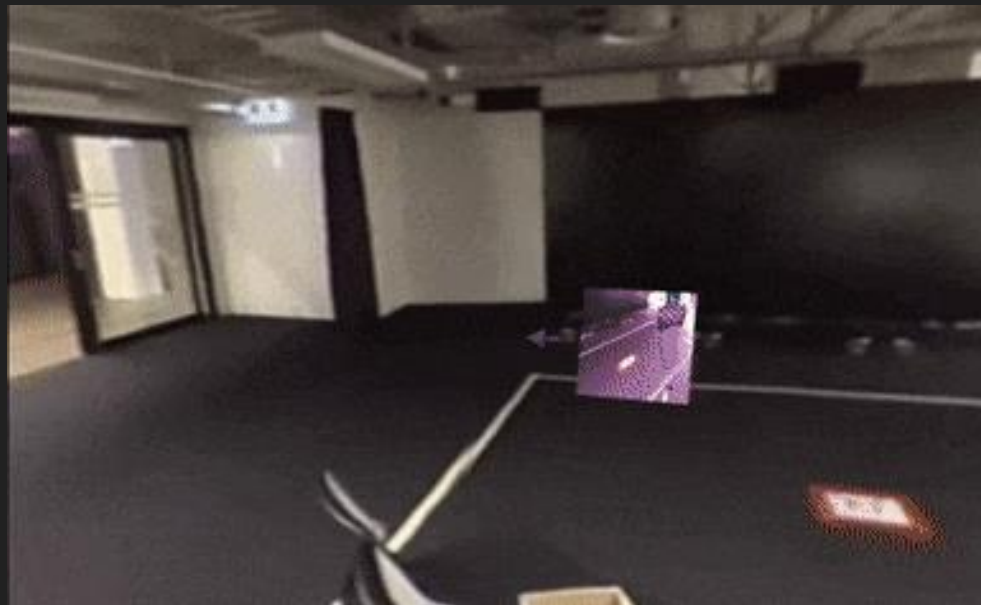


3. Spatial Indication



Visualization and Interaction

3. Spatial Indication



Vicarious: Research Questions

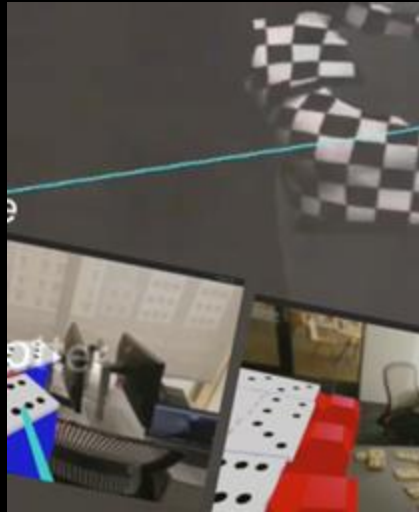
viewpoint perspective

RQ4: How can users' **viewpoints** be managed effectively to support various types of collaborative tasks in the Mixed Reality Collaboration space?

**No Selection
(NS)**



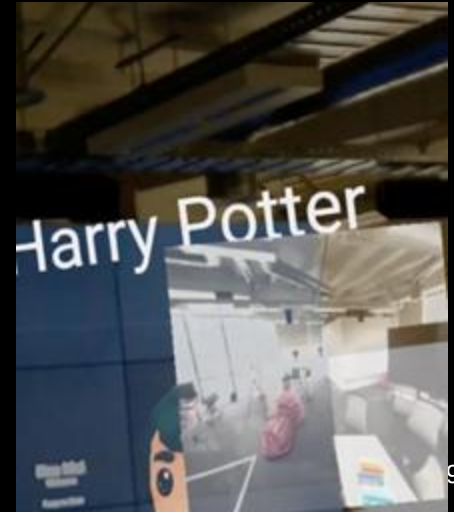
**Manual
Selection (MS)**



**Guided
Selection (GS)**



**Automatic
Selection (AS)**



Vicarious: Research Questions

viewpoint perspective

RQ4: How can users' **viewpoints be managed effectively** to support various types of collaborative tasks in the Mixed Reality Collaboration space?

multi-perspective collaboration

RQ5: How can **multi-perspective collaboration be effectively** achieved for supporting collaborative tasks?

RQ5.1 Would context-aware viewpoint selection increase the **sense of presence** in remote collaboration?

RQ5.2 Would context-aware viewpoint selection improve the **task performance** compared to using no-view selection?

Experimental Setup

4x2 mixed factorial design

between-subjects factor:
local vs. remote

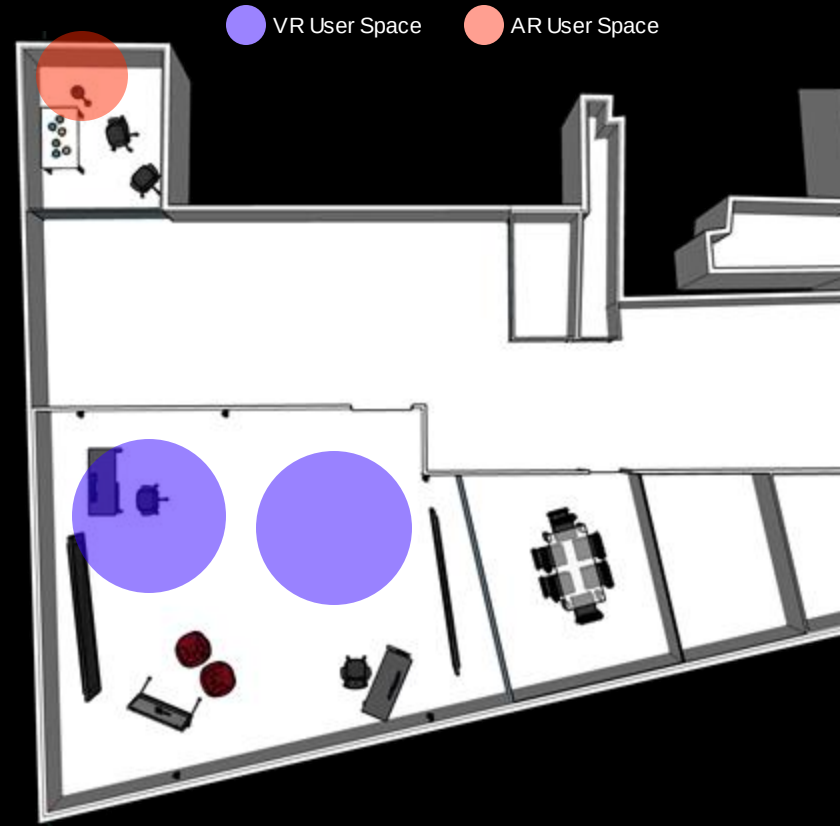
within-subjects factor:
NS vs. MS vs. GS vs. AS

27 participants

18—55 years

$\mu = 28.13$

$\sigma = 7.32$



Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness

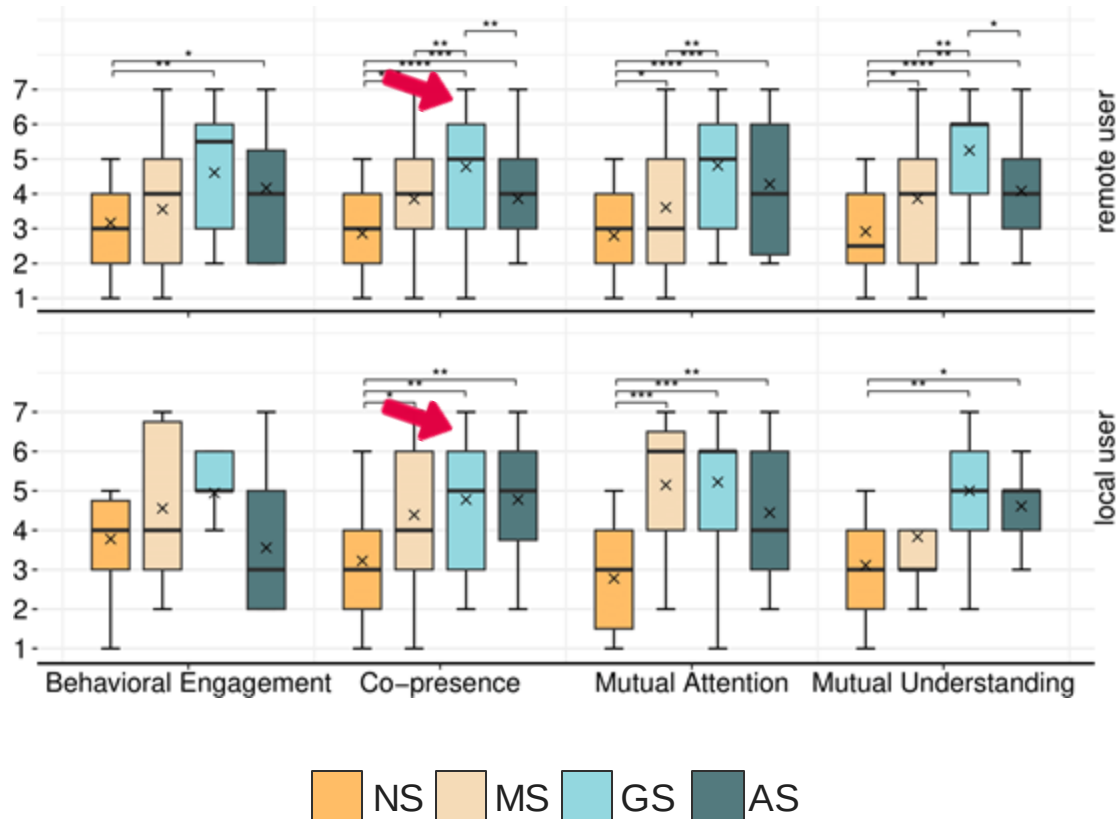
Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness



Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

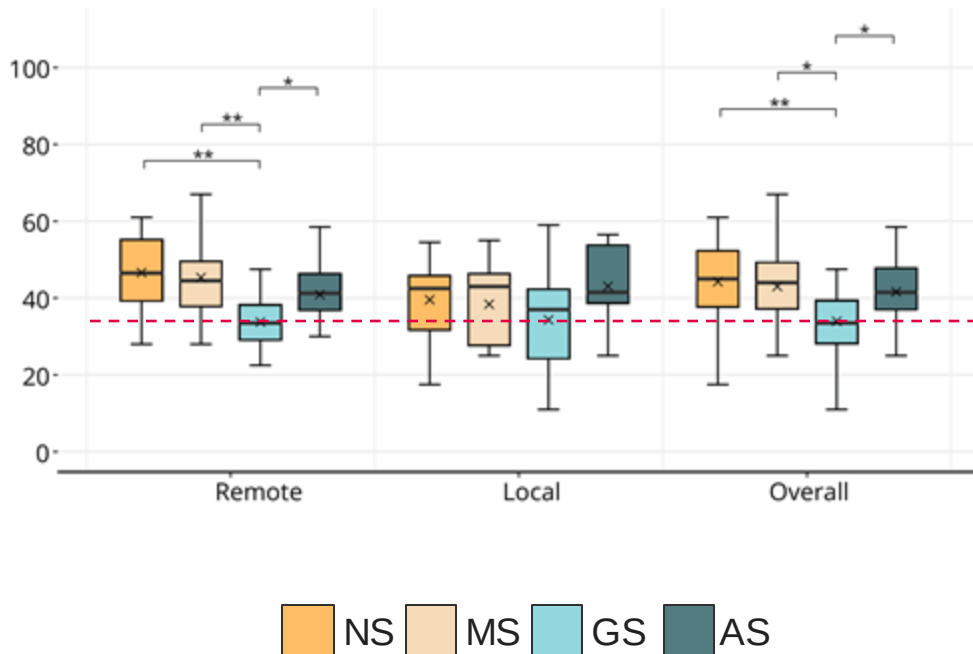
Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness

GS

Local: M = 32.31, SD = 23.96

Remote: M = 34.76, SD = 25.41



Results

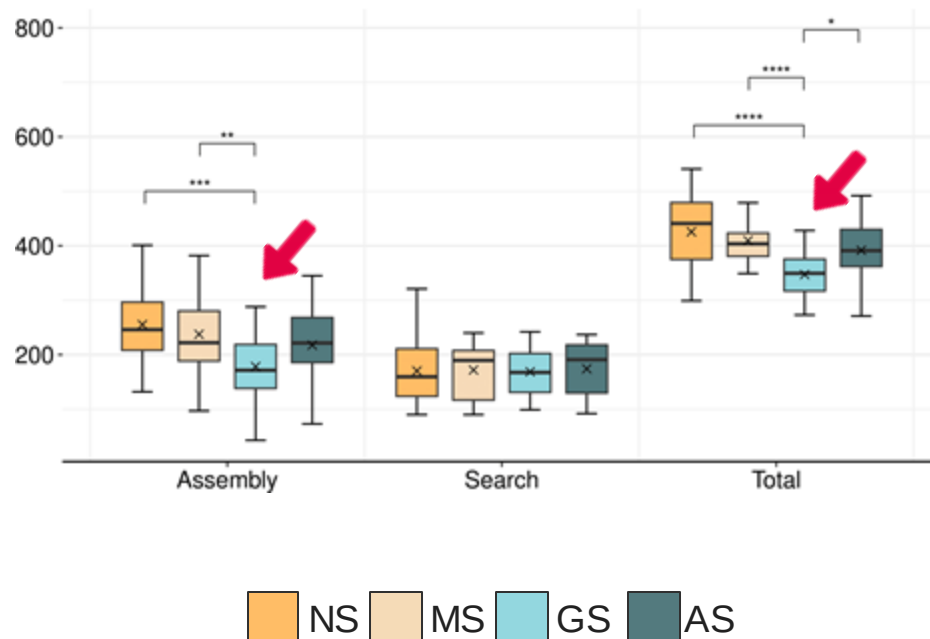
Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness

GS: $M = 393.46$, $SD = 60.32$



Results

Statistically Significant

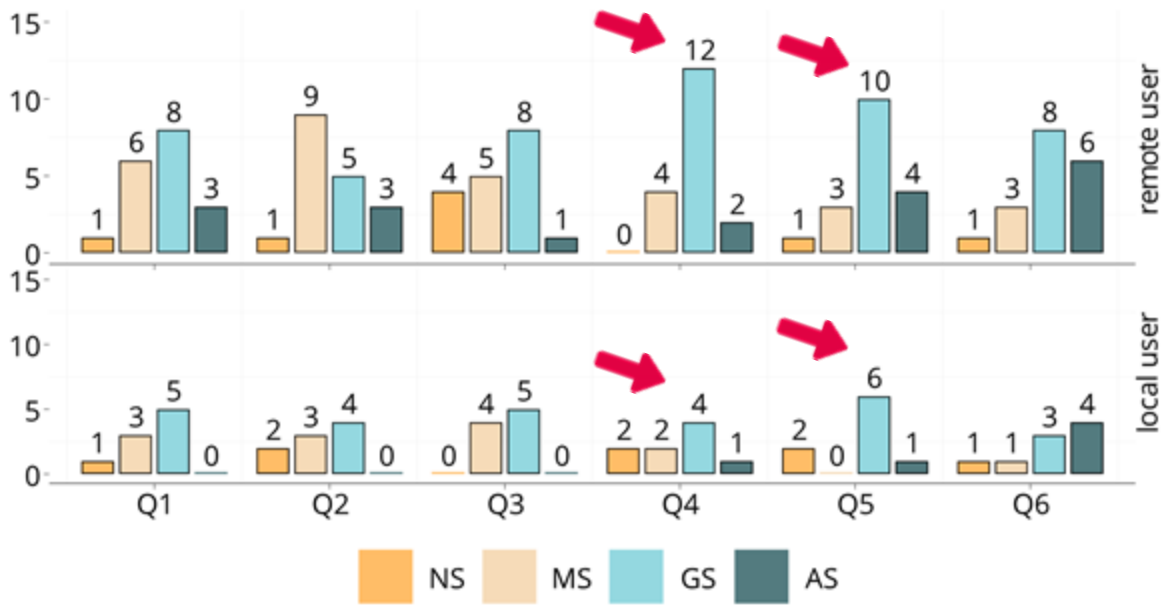
- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness

Q4 Which condition did you find most effective for communicating with your partner?

Q5 Which condition did you find most helpful for completing the task?



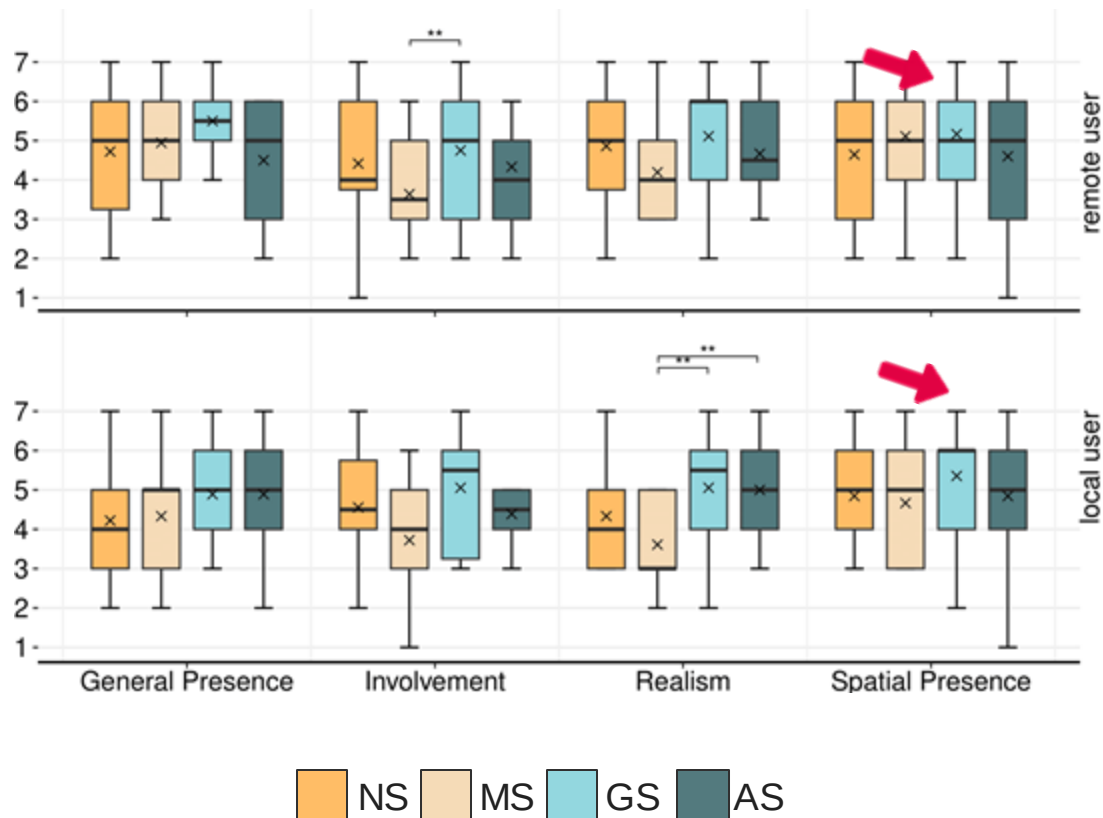
Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness



Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

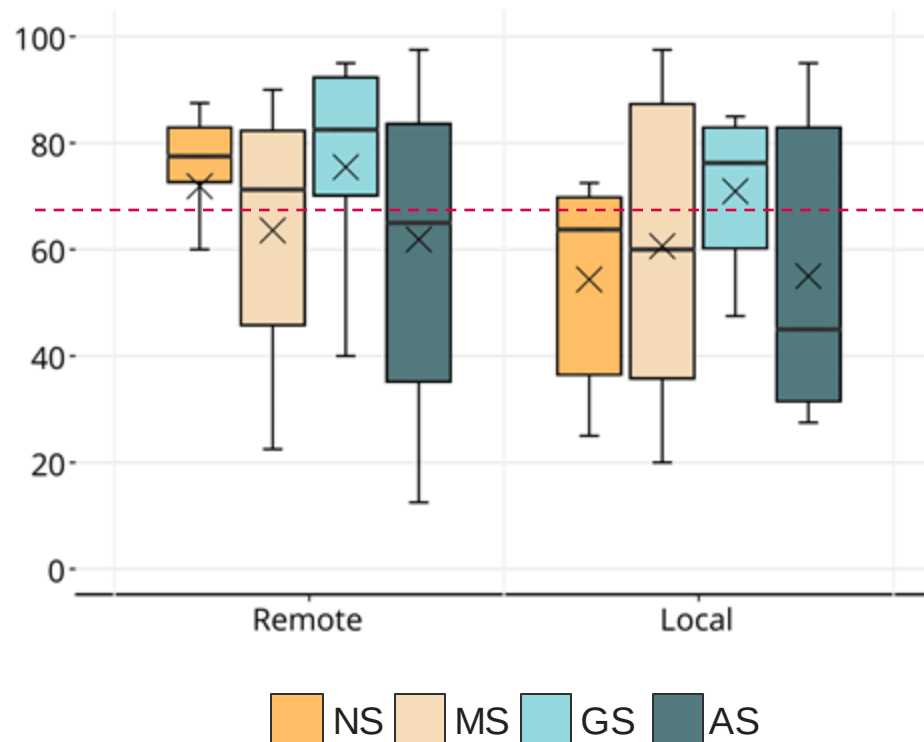
Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness

GS

Local: M = 68.61, SD = 15.87

Remote: M = 74.58, SD = 19.82



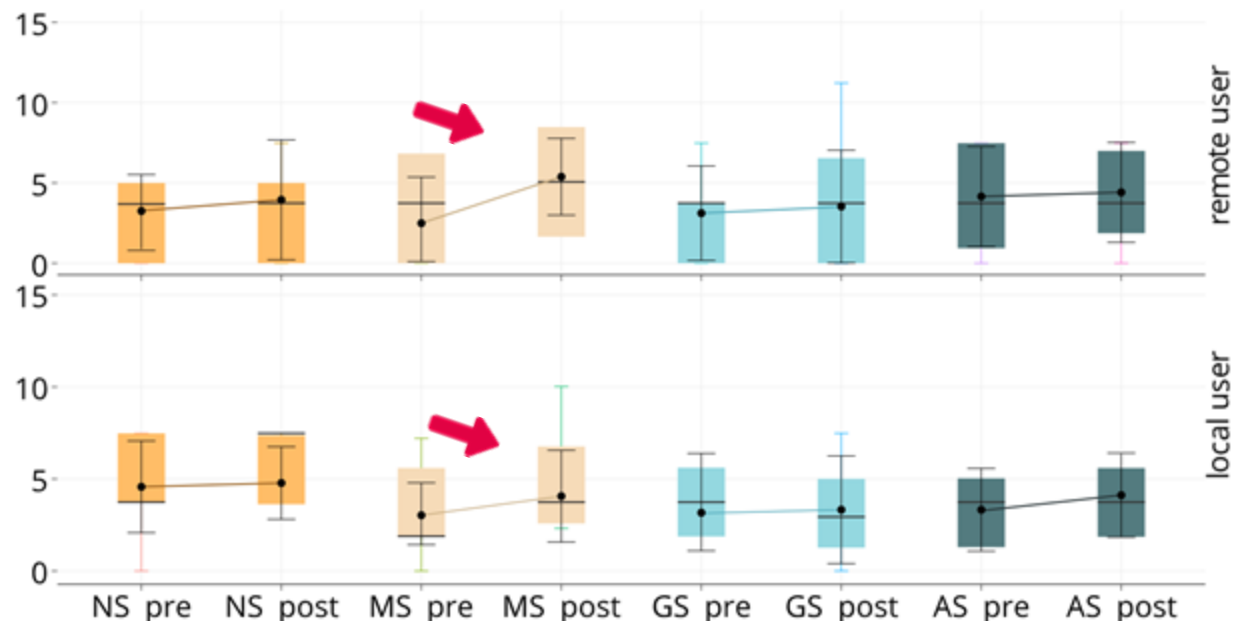
Results

Statistically Significant

- Social Presence
- Task Workload
- Task Completion Time
- Preference

Not Statistically Significant

- Spatial Presence
- System Usability
- Simulator Sickness



Summary of Vicarious

- **Context-aware viewpoint selection method**
 - simplifies and automates multiple viewpoints.
 - finds optimal viewpoints based on audio-visual cues.
- **User evaluation**
 - 27 participants under four distinct conditions
 - asymmetric multiuser AR-VR setup
- **Findings and design implications**
 - improved task space understanding and performance
 - reduced cognitive load
 - GS was the most preferred

Why work remotely when
you can work *Vicariously*?



F. Zaman, C. Anslow, T. Rhee.
**Vicarious: Context-aware
Viewpoints Selection for Mixed
Reality Collaboration.**
ACM VRST, 2023.

Summary of Contributions

RQ1: teleportation

- Rich real-time telepresence experiences can feasibly be created
- MRMAC induces significantly more spatial presence than 2D and 360 video alone
- These experiences still pose a significant engineering challenge
- Opened a lot of future research opportunities

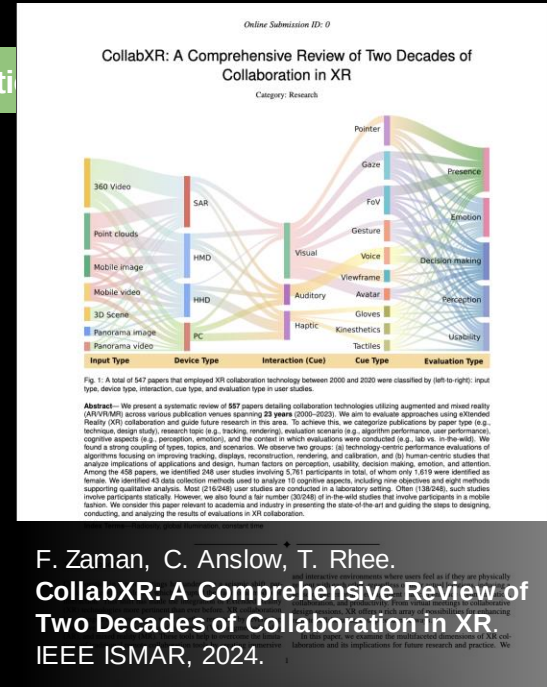
RQ4: viewpoint perspective

- Optimal viewpoint selection improved users' task space understanding
- Task performance and reduced cognitive load.
- Users preferred the intervention

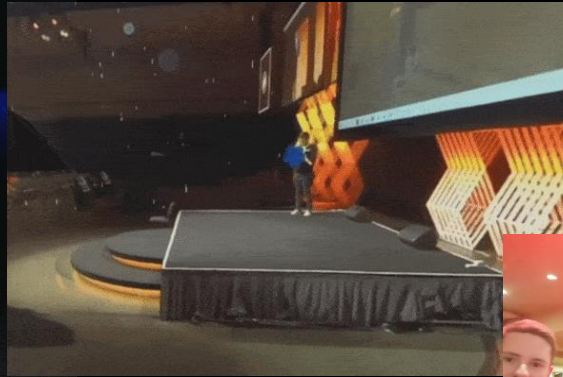
RQ2: representation

RQ3: interaction

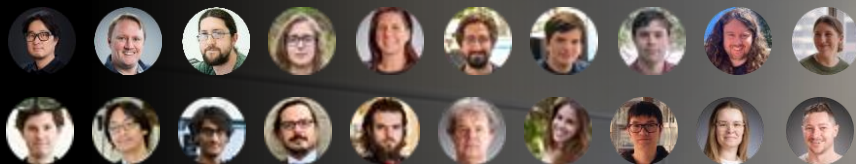
RQ5: multi-perspective collaboration



Research Journey



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